Grade 10 Term 3 — Exam Solutions (C4)

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Evaluated criterion

C4. Interprets the concept of expected value, variance, and standard deviation of a probability function.

Interpretation focus used in this solution set:

- The **expected value** is interpreted as the horizontal *balance point/center* of the graph.
- The **standard deviation** is interpreted as the typical horizontal *spread* around the center.
- **Variance** is interpreted as spread squared; larger variance means a more dispersed graph.
- The **shape** (flat, slanted, triangular, bell-shaped) helps explain where probability mass is concentrated.

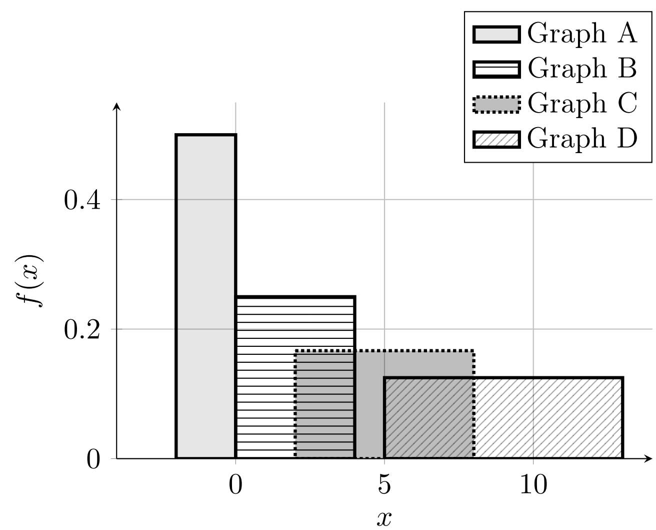
Uniform Distribution

For a uniform distribution $U[a, b]$, the center is the midpoint of the interval, and the spread depends only on the interval width $b - a$, so

$$\mu = \frac{a+b}{2} \qquad \sigma = \frac{b-a}{\sqrt{12}}$$

1 1. (1) Uniform: match graphs with their means

Four uniform densities are shown. The distributions have different spreads, but the task is only to match each graph with its mean.



Match each graph with one mean option. Some options will not be used.

- Mean options: $\{-1,\ 0,\ 2,\ 5,\ 9,\ 12\}$

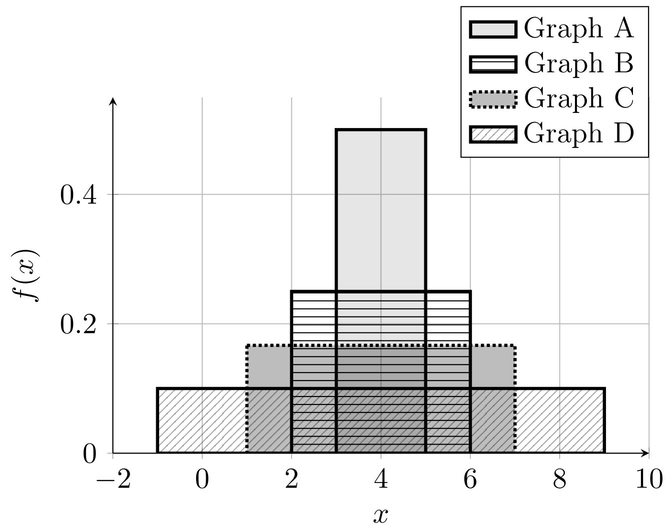
Solution

Final match.

- Graph A $\to \mu = -1$
- Graph B $\to \mu = 2$
- Graph C $\to \mu = 5$
- Graph D $\to \mu = 9$

2 2. (1) Uniform: match graph with their spread

Four uniform densities are shown. All four distributions have the same expected value, so only the standard deviation changes.



Match each graph with one standard deviation option. Some options will not be used.

- Standard deviation options (approximately): $\{0.58,\ 1.15,\ 1.73,\ 2.89\}$

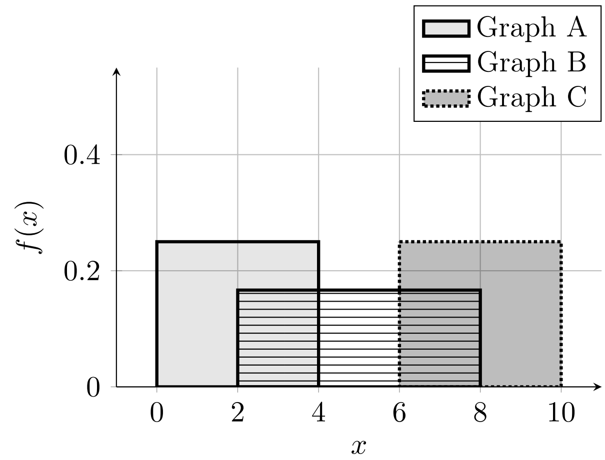
Solution

Final match.

- Graph A $\rightarrow \sigma \approx 0.58$
- Graph B $\rightarrow \sigma \approx 1.15$
- Graph C $\rightarrow \sigma \approx 1.73$
- Graph D $\rightarrow \sigma \approx 2.89$

3 3. (1) Match graph, mean, and spread for three uniform distributions

Three uniform densities are shown.



Match each graph with one option from each list.

- Mean options: $\{0,\ 2,\ 5,\ 7,\ 8, 9\}$
- Standard deviation options (approximately): $\{0.15,\ 1.15,\ 1.15,\ 1.73\}$

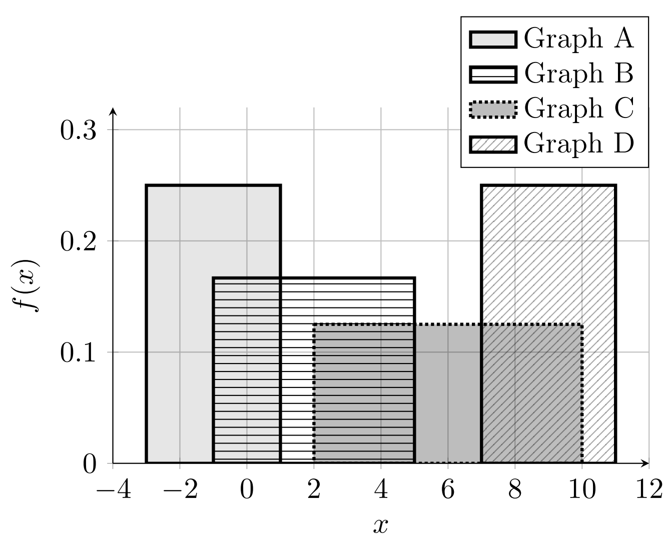
Solution

Final match.

- Graph A $\rightarrow \mu = 2$, $\sigma \approx 1.15$
- Graph B $\rightarrow \mu = 5$, $\sigma \approx 1.73$
- Graph C $\rightarrow \mu = 8$, $\sigma \approx 1.15$

4 4. (2) Match graph, mean, and spread for four uniform distributions

Four uniform densities are shown.



Match each graph with one option from each list. Some options will not be used.

- Mean options: $\{-3,\ -1,\ 2,\ 6,\ 9,\ 10\}$
- Standard deviation options (approximately): $\{0.58,\ 1.15,\ 1.15,\ 1.73,\ 2.31\}$

Solution

Final match.

- Graph A $\to \mu = -1$, $\sigma \approx 1.15$
- Graph B $\to \mu = 2$, $\sigma \approx 1.73$
- Graph C $\to \mu = 6$, $\sigma \approx 2.31$
- Graph D $\to \mu = 9$, $\sigma \approx 1.15$

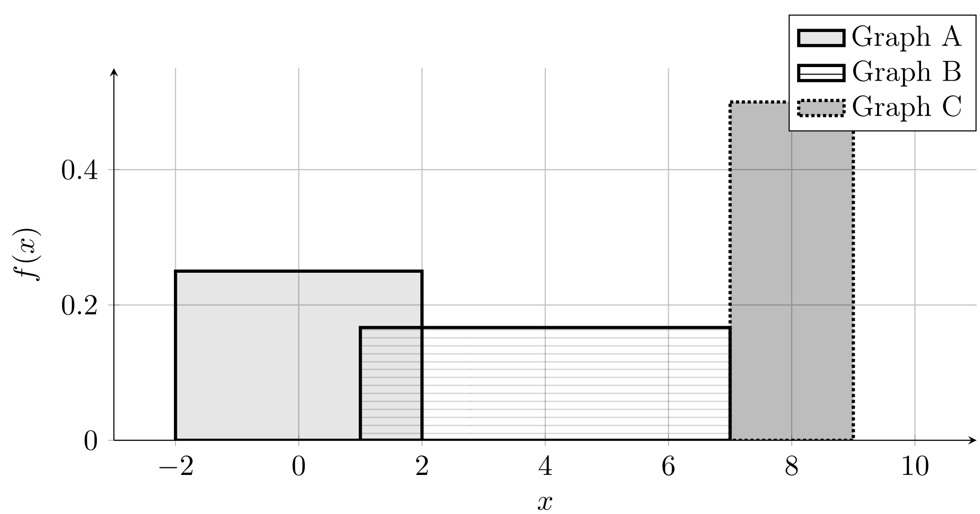
5 5. (2) Uniform: draw three uniform distributions on the same graph

Draw the following three uniform distributions on the same set of axes.

- Graph A: mean $\mu = 0$ and standard deviation $\sigma \approx 1.15$
- Graph B: mean $\mu = 4$ and standard deviation $\sigma \approx 1.73$
- Graph C: mean $\mu = 8$ and standard deviation $\sigma \approx 0.58$

Use the same coordinate plane for all three distributions. Clearly distinguish the graphs in the picture.

Solution



6 6. (2) Uniform: draw four uniform distributions on the same graph

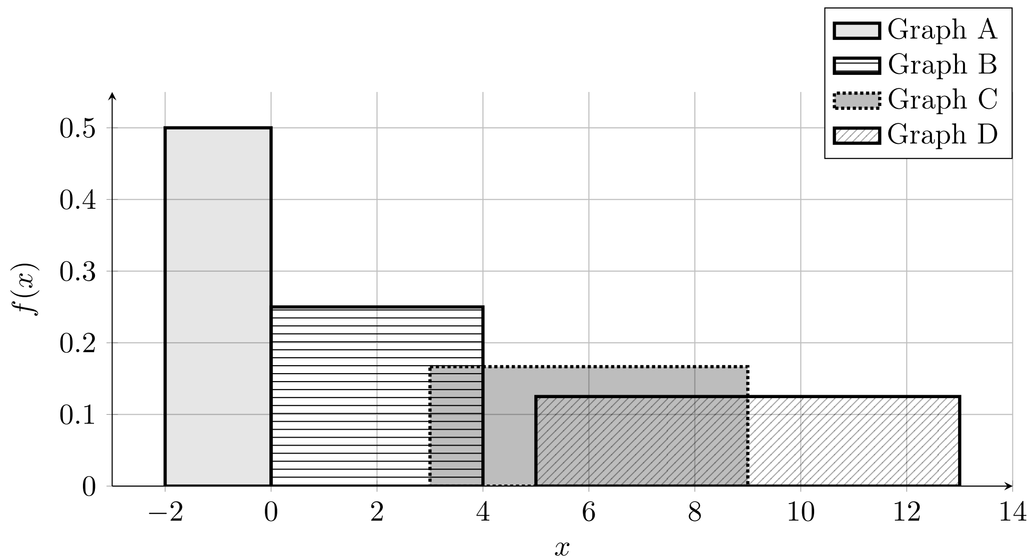
Draw the following four uniform distributions on the same set of axes.

- Graph A: mean $\mu = -1$ and standard deviation $\sigma \approx 0.58$

- Graph B: mean $\mu = 2$ and standard deviation $\sigma \approx 1.15$
- Graph C: mean $\mu = 6$ and standard deviation $\sigma \approx 1.73$
- Graph D: mean $\mu = 9$ and standard deviation $\sigma \approx 2.31$

Use the same coordinate plane for all four distributions. Clearly distinguish the graphs in the picture.

Solution



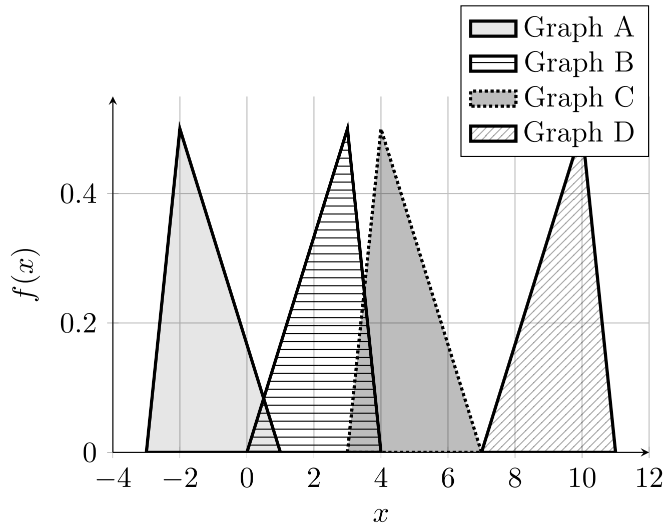
Triangular Distribution

For a triangular distribution with left endpoint a, mode c, and right endpoint b, the center depends on the three values a, c, and b, while the spread depends on how far apart these values are, so

$$\mu = \frac{a+b+c}{3} \qquad \sigma = \sqrt{\frac{a^2+b^2+c^2-ab-ac-bc}{18}}$$

7 7. (1) Triangular: same spread, different mean

Four triangular densities are shown. All four distributions have the same standard deviation, so only the expected value changes.



Match each graph with one mean option. Some options will not be used.

- Mean options: $\left\{-\frac{4}{3},\ 0,\ \frac{7}{3},\ \frac{14}{3},\ 8,\ \frac{28}{3},\ 11\right\}$

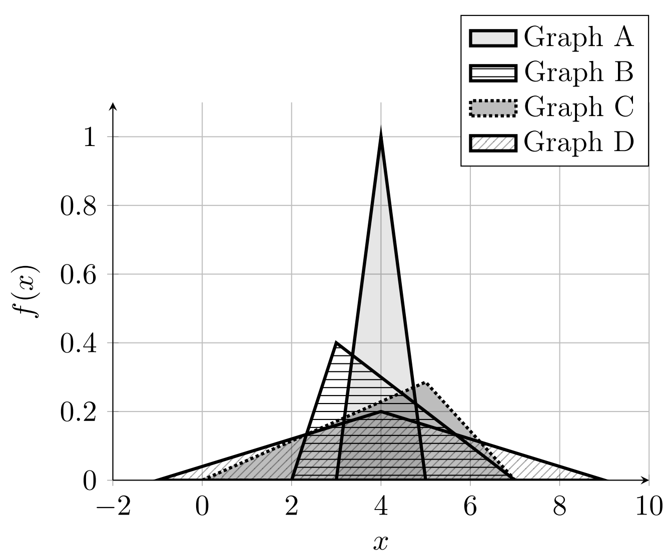
Solution

Final match.

- Graph A $\rightarrow \mu = -\frac{4}{3}$
- Graph B $\rightarrow \mu = \frac{7}{3}$
- Graph C $\rightarrow \mu = \frac{14}{3}$
- Graph D $\rightarrow \mu = \frac{28}{3}$

8 8. (1) Triangular: same mean, different spread

Four triangular densities are shown. All four distributions have the same expected value, so only the standard deviation changes.



Match each graph with one standard deviation option. Some options will not be used.

- Standard deviation options (approximately): $\{0.41,\ 1.08,\ 1.47,\ 2.04\}$

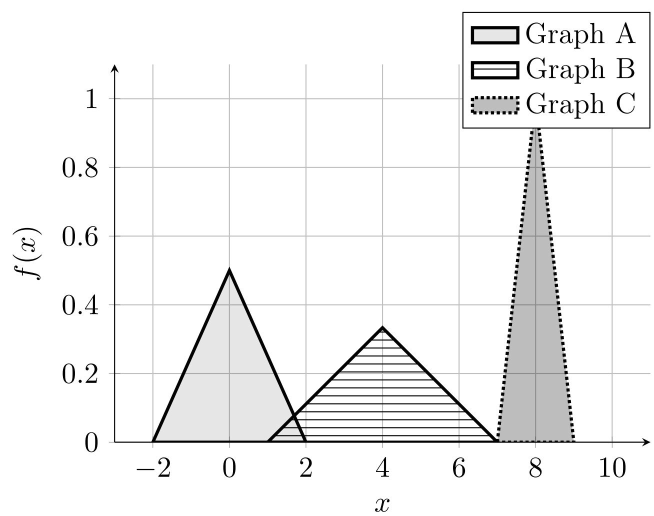
Solution

Final match.

- Graph A $\to \sigma \approx 0.41$
- Graph B $\to \sigma \approx 1.08$
- Graph C $\to \sigma \approx 1.47$
- Graph D $\to \sigma \approx 2.04$

9 9. (1) Match graph, mean, and spread for three triangular distributions

Three triangular densities are shown.



Match each graph with one option from each list.

- Mean options: $\{-2,\ 0,\ 2,\ 4,\ 6,\ 8\}$
- Standard deviation options (approximately): $\{0.41,\ 0.82,\ 1.22\}$

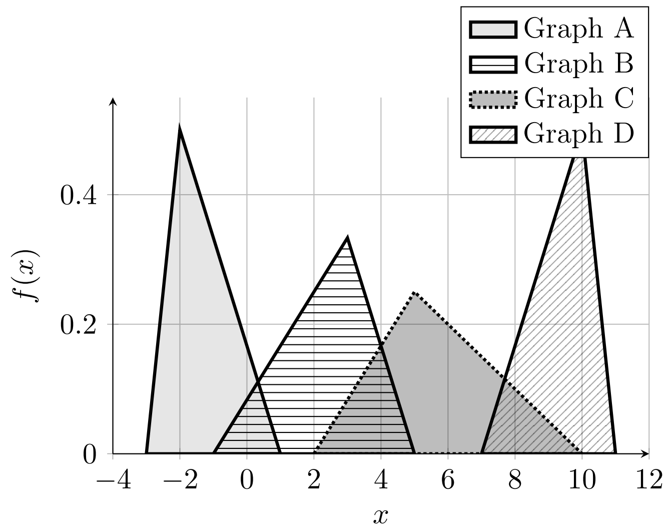
Solution

Final match.

- Graph A $\to \mu = 0$, $\sigma \approx 0.82$
- Graph B $\to \mu = 4$, $\sigma \approx 1.22$
- Graph C $\to \mu = 8$, $\sigma \approx 0.41$

10 10. (2) Match graph, mean, and spread for four triangular distributions

Four triangular densities are shown.



Match each graph with one option from each list. Some options will not be used.

- Mean options: $\left\{-\frac{4}{3},\ 0,\ \frac{7}{3},\ \frac{17}{3},\ \frac{28}{3},\ \frac{34}{3}\right\}$
- Standard deviation options (approximately): $\{0.41,\ 0.65,\ 0.85,\ 0.85,\ 1.25,\ 1.65\}$

Solution

Final match.

- Graph A $\to \mu = -\frac{4}{3}$, $\sigma \approx 0.85$
- Graph B $\to \mu = \frac{7}{3}$, $\sigma \approx 1.25$
- Graph C $\to \mu = \frac{17}{3}$, $\sigma \approx 1.65$
- Graph D $\to \mu = \frac{28}{3}$, $\sigma \approx 0.85$

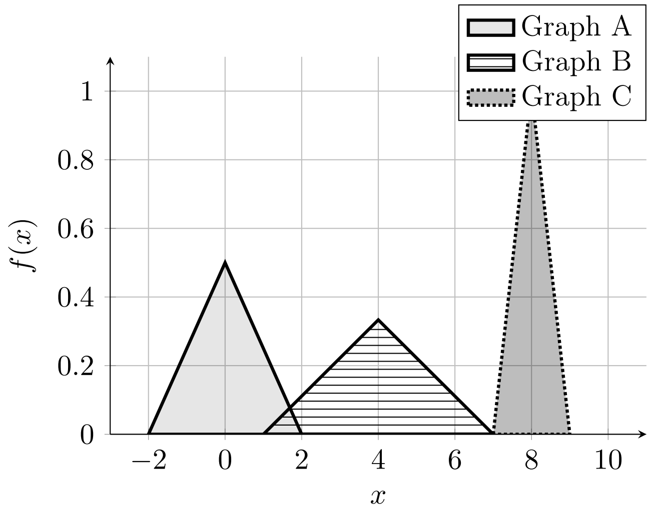
11 11. (2) Triangular: draw three triangular distributions on the same graph

Draw the following three triangular distributions on the same set of axes.

- Graph A: mean $\mu = 0$ and standard deviation $\sigma \approx 0.82$
- Graph B: mean $\mu = 4$ and standard deviation $\sigma \approx 1.22$
- Graph C: mean $\mu = 8$ and standard deviation $\sigma \approx 0.41$

Use the same coordinate plane for all three distributions. Clearly distinguish the graphs in the picture.

Solution



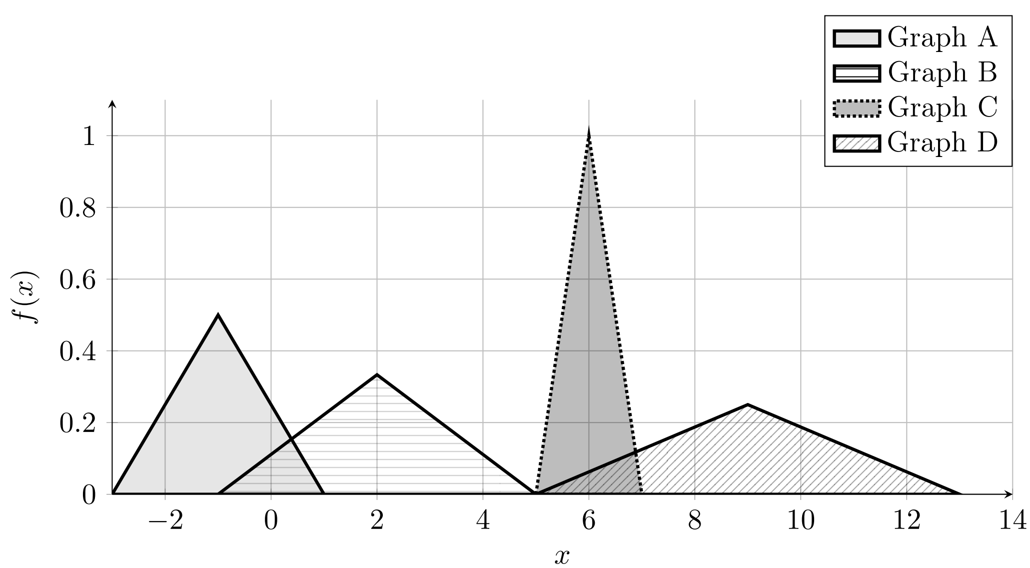
12 12. (2) Triangular: draw four triangular distributions on the same graph

Draw the following four centered triangular distributions on the same set of axes.

- Graph A: mean $\mu = -1$ and standard deviation $\sigma \approx 0.82$
- Graph B: mean $\mu = 2$ and standard deviation $\sigma \approx 1.22$
- Graph C: mean $\mu = 6$ and standard deviation $\sigma \approx 0.41$
- Graph D: mean $\mu = 9$ and standard deviation $\sigma \approx 1.63$

Use the same coordinate plane for all four distributions. Clearly distinguish the graphs in the picture.

Solution



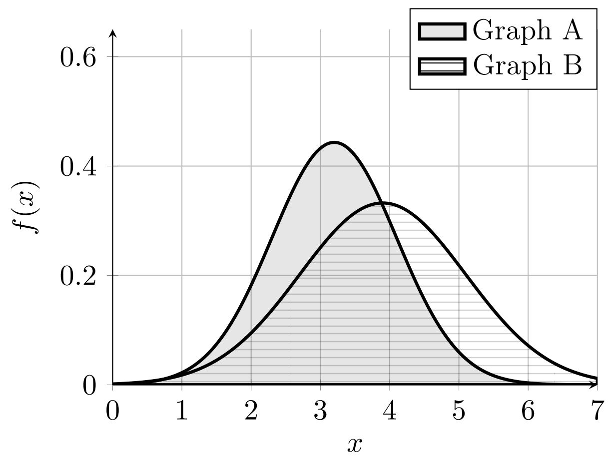
Normal Distribution

For a normal distribution $N(\mu, \sigma^2)$, the center is the parameter μ, and the spread is the standard deviation σ, so

$$E[X] = \mu \qquad \mathrm{SD}(X) = \sigma$$

13 13. (1) Normal I: match two graphs with their mean and standard deviation

Two normal densities are shown. Both the expected value and the standard deviation change from one graph to the other.



Match each graph with one mean option and one standard deviation option. Some options will not be used.

- Mean options: $\{2.2,\ 3.2,\ 3.9,\ 4.9,\ 5.2\}$
- Standard deviation options: $\{0.9,\ 1.2\}$

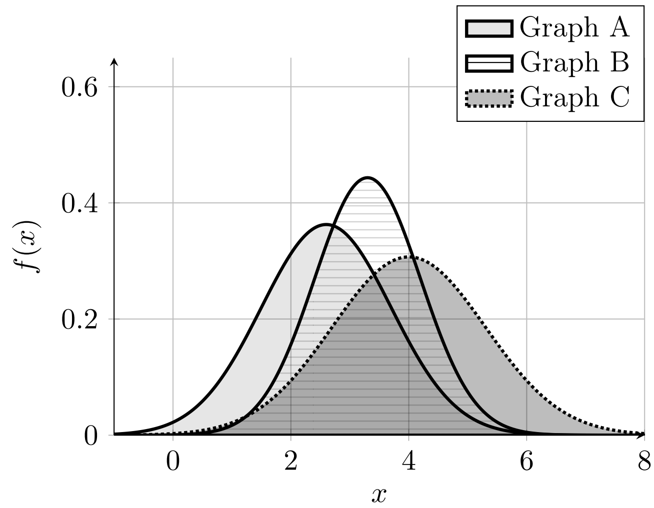
Solution

Final match.

- Graph A $\to$ $\mu = 3.2$, $\sigma = 0.9$
- Graph B $\to$ $\mu = 3.9$, $\sigma = 1.2$

14 14. (1) Normal II: match three graphs with their mean and standard deviation

Three normal densities are shown. Each graph has a different expected value and a different standard deviation.



Match each graph with one mean option and one standard deviation option. Some options will not be used.

- Mean options: $\{2.0,\ 2.6,\ 3.0,\ 3.3,\ 4.0,\ 4.9,\ 5.5\}$
- Standard deviation options: $\{0.9,\ 1.1,\ 1.3\}$

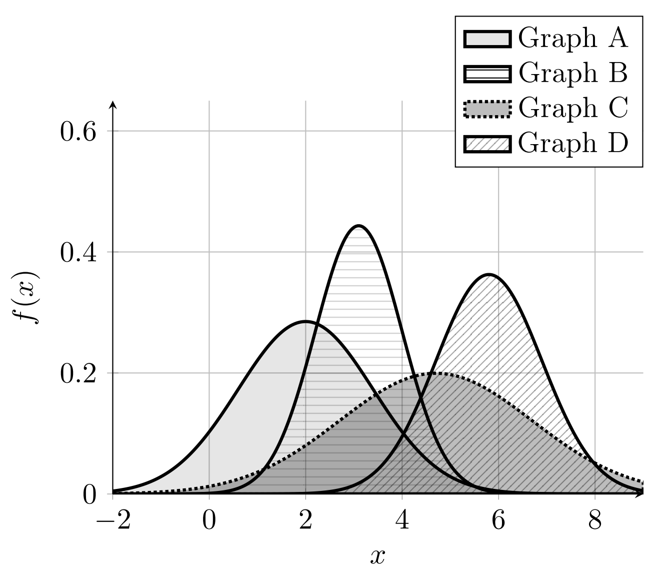
Solution

Final match.

- Graph A $\rightarrow$ $\mu = 2.6$, $\sigma = 1.1$
- Graph B $\rightarrow$ $\mu = 3.3$, $\sigma = 0.9$
- Graph C $\rightarrow$ $\mu = 4.0$, $\sigma = 1.3$

15 15 (1). Normal III: match four graphs with their mean and standard deviation

Four normal densities are shown. Each graph has a different expected value and a different standard deviation.



Match each graph with one mean option and one standard deviation option. Some options will not be used.

- Mean options: $\{1.2,\ 2.0,\ 3.1,\ 3.9,\ 4.7,\ 5.8,\ 6.8\}$
- Standard deviation options: $\{0.9,\ 1.1,\ 1.4,\ 2.0\}$

Solution

Final match.

- Graph A $\rightarrow \mu = 2.0$, $\sigma = 1.4$
- Graph B $\rightarrow \mu = 3.1$, $\sigma = 0.9$
- Graph C $\rightarrow \mu = 4.7$, $\sigma = 2.0$
- Graph D $\rightarrow \mu = 5.8$, $\sigma = 1.1$

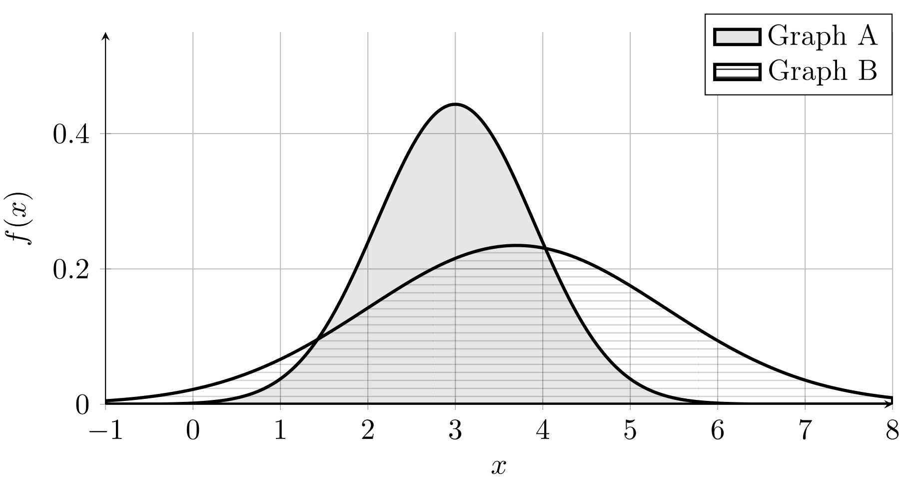
16 16. (2) Normal IV: draw two normal distributions on the same graph

Draw the following two normal distributions on the same set of axes.

- Graph A: $X \sim N(3.0, 0.9^2)$
- Graph B: $Y \sim N(3.7, 1.7^2)$

Use the same coordinate plane for both distributions. Clearly distinguish the graphs in the picture.

Solution



17 17. (2) Normal V: draw three normal distributions on the same graph

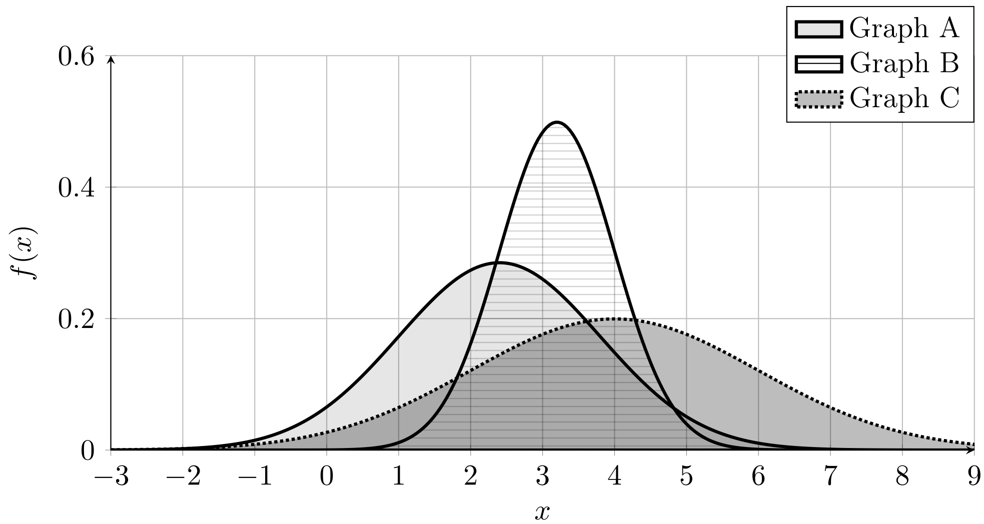
Draw the following three normal distributions on the same set of axes.

- Graph A: $X \sim N(2.4, 1.4^2)$

- Graph B: $Y \sim N(3.2, 0.8^2)$
- Graph C: $Z \sim N(4.0, 2.0^2)$

Use the same coordinate plane for all three distributions. Clearly distinguish the graphs in the picture.

Solution



18 18. (3) Normal VI: draw four normal distributions on the same graph

Draw the following four normal distributions on the same set of axes.

- Graph A: $X \sim N(1.0, 1.2^2)$
- Graph B: $Y \sim N(3.0, 0.7^2)$
- Graph C: $Z \sim N(5.0, 1.6^2)$
- Graph D: $W \sim N(7.0, 1.0^2)$

Use the same coordinate plane for all four distributions. Clearly distinguish the graphs in the picture.

Solution